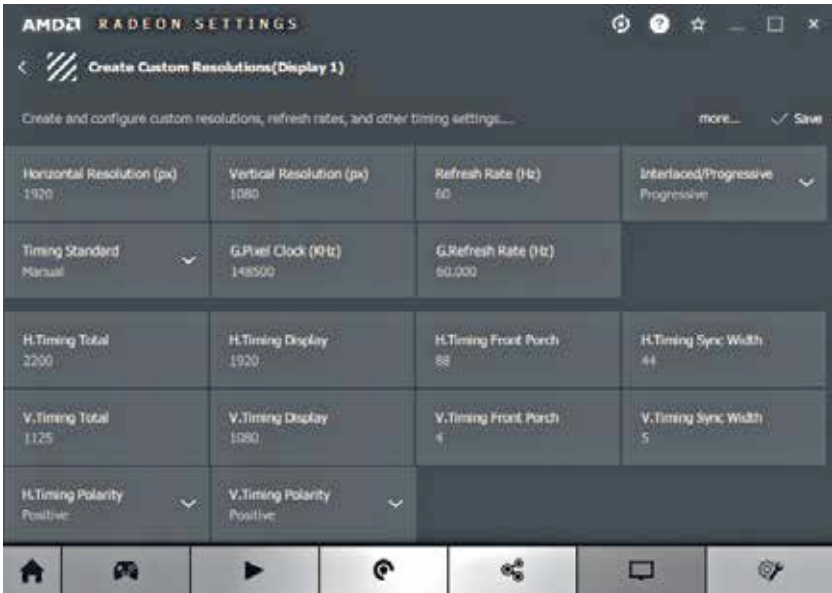


OVERCLOCKING YOUR MONITOR

Things to know

Your monitor's native refresh rate isn't a limit but rather a stable and safe number that will ensure longevity. It's usually set by the manufacturer and users will be able to squeeze out around 10-15 Hz more than the specified number. So, if your monitor's specifications mention a native refresh rate of 60Hz, you might be able to hit 75Hz or even up to 85Hz in some cases. It all depends on the hardware used in the monitor. The difference isn't significant enough but you will still be able to churn out a higher refresh rate than what the manufacturer wanted. However, if the display already has a high refresh rate then don't expect to be able to scale too much.

When you're overclocking, you also need to consider the extra bandwidth required. However, since the overclocked refresh rate isn't going to be massive, your existing cable should be capable enough to handle it. To be on a safer side, it's best to use a DisplayPort. Finally, overclocking your monitor might void your warranty so if your display blows out later for different reasons, you will have to shell out the money. Increasing the refresh rate won't really damage your display but it might lower the durability in the long run.



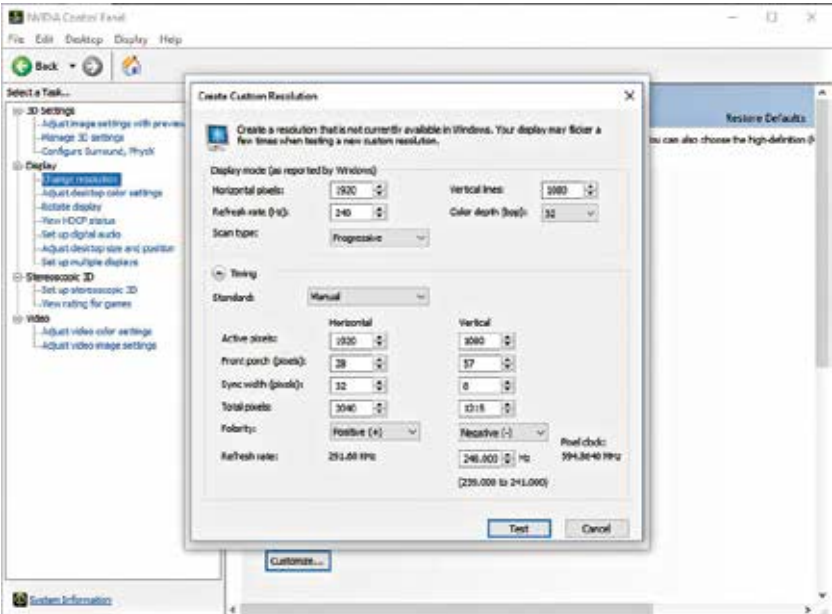
AMD's Adrenalin 2019 software suite allows custom resolutions

Increasing the refresh rate

It doesn't matter whether you're using an integrated graphics card or a discrete one, there are different ways to overclock your display's refresh rate. The process is quite simple and it can be done using the default software provided by the graphics card manufacturers. Starting off with AMD, their Adrenalin 2019 software utility packs in a lot of features, one of which enables you to overclock your monitor. Head over to the Display tab and enable Virtual Super Resolution. Once enabled, you can create Custom Resolutions. Click on Create and you'll

be taken to a new tab where you can edit the values of several parameters. We are only interested in the refresh rate and you will see the native value already entered. Rather than directly adding 75Hz, it's best to increment by 5Hz to check for stability. After you enter a value and save the custom resolution, create two or three more profiles with different values. Next, navigate to your system's Display Settings by right-clicking on your desktop. Scroll down to 'Advanced display settings' and then 'Display adapter properties'. Go to the Monitor tab and click on the dropdown list and you will find all the different refresh rate values listed. Select them one by one and check whether the display reappears. In our case, we were able to reach a maximum of 75Hz. When 80Hz was enabled, the screen went blank and returned to the previous setting after 15 seconds.

Coming to NVIDIA cards, right-click on your desktop and head over to the NVIDIA Control Panel. Under the Display tab, click on Change Resolution then Customise and finally Create Custom Resolution. On the new window, increase the refresh rate in increments and verify whether it successfully enables using the Test button on the bottom. If the test is successful, you won't get the black screen and it ask you whether you want to apply the new refresh rate. Click on yes and enjoy a smoother display. 



Overclock your monitor using NVIDIA Control Panel by creating a custom resolution profile